CD-Player

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Chapter 1

Preparation

1.1 Part One

a)

With the equation of motion

$$m\ddot{x} + Kx = f, \tag{1.1}$$

and subsequent Laplace transform (given zero initial conditions):

$$ms^2X(s) + KX(s) = F(s), \tag{1.2}$$

the transfer function form

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + K}, \tag{1.3}$$

is derived. For a negligible spring constant($K \approx 0$), the undamped system is derived:

$$G(s) = \frac{1}{ms^2}. \tag{1.4}$$

b)

We want to derive the transfer function from the driving voltage $V(s)$ to the measured error. The complete loop includes the plant $P(s)$, the mechanical system $G(s)$, and the sensor gain $K(s)$, such that the overall transfer function becomes:

$$P(s)G(s)K(s). \tag{1.5}$$

Neglecting the actuator's inductance ($L = 0$), the Lorentz force becomes directly proportional to the current:

$$F(s) = K_M I(s). \tag{1.6}$$

With the current given by Ohm's law,

$$I(s) = \frac{V(s)}{R}, \tag{1.7}$$

we substitute and obtain the plant transfer function:

$$P(s) = \frac{F(s)}{V(s)} = \frac{K_M}{R}. \tag{1.8}$$

Using the mechanical system transfer function $G(s)$ from Exercise a (equation (1.3)), and a sensor system modeled as a constant gain:

$$K(s) = \pm K_g, \tag{1.9}$$

the complete system becomes:

$$\frac{X_{\text{measured}}(s)}{V(s)} = \left(\frac{K_M}{R}\right)\left(\frac{1}{ms^2 + K}\right) K(s). \tag{1.10}$$

c)

We assume the spring constant is negligible ($K = 0$), so the system behaves as a floating mass. From part (b), the overall transfer function from the input voltage $V(s)$ to the measured output is:

$$P(s)G(s)K(s) = \frac{K_M K_g}{R(ms^2 + K)}. \tag{1.11}$$

Substituting $K = 0$ simplifies the expression to:

$$P(s)G(s)K(s) = \frac{K_M K_g}{Rms^2}. \tag{1.12}$$

This matches the canonical form of a floating mass system:

$$\frac{a}{s^2}, \tag{1.13}$$

from which we identify the constant:

$$a = \frac{K_M K_g}{Rm}. \tag{1.14}$$

d)

The output voltage of the circuit in area B is derived in relation to the input voltage is derived by combining knowledge of the ideal OP amps with

Kirchhoff's Current Law.

$$U_1 = -U_{\text{in}} \cdot \frac{R_8}{R_7} \tag{1.15}$$

$$U_2 = -U_{\text{in}} \cdot \frac{R_{10}}{Z_1} \tag{1.16}$$

$$Z_1 = \frac{R_9 \cdot (j\omega C_1 + 1)}{j\omega C_1} \tag{1.17}$$

$$U_2 = -U_{\text{in}} \cdot \frac{R_{10} j\omega C_1}{R_9 j\omega C_1 + 1} \tag{1.18}$$

$$U_3 = -\frac{Z_3}{R_{11}} \cdot U_{\text{in}} \tag{1.19}$$

$$Z_3 = \frac{R_{19}}{1 + j\omega C_2 R_{19}} \tag{1.20}$$

$$U_3 = -\frac{R_{19}}{R_{11}\left(1 + j\omega C_2 R_{19}\right)} \cdot U_{\text{in}} \tag{1.21}$$

$$I_1 = \frac{U_1}{R_{12}} \tag{1.22}$$

$$I_2 = \frac{U_2}{R_{13}} = -U_{\text{in}} \cdot \frac{R_{10} \cdot j\omega C_1}{R_{13} \cdot (R_9 j\omega C_1 + 1)} \tag{1.23}$$

$$I_3 = \frac{U_3}{R_{14}} = -U_{\text{in}} \cdot \frac{R_{19}}{R_{11}(1 + j\omega C_2 R_{19})} \cdot \frac{1}{R_{14}} \tag{1.24}$$

$$U_{\text{out}} = -(I_1 + I_2 + I_3) \cdot R_{15} \tag{1.25}$$

$$U_{\text{out}} = U_{\text{in}} \cdot \left(\frac{R_{15} R_8}{R_7 R_{12}} + \frac{R_{10} j\omega C_1 R_{15}}{(R_9 j\omega C_1 + 1) R_{13}} + \frac{R_{19} R_{15}}{R_{11} R_{14} (1 + j\omega C_2 R_{19})} \right) \tag{1.26}$$

$$\Rightarrow \quad C(s) = \frac{U_{\text{out}}}{U_{\text{in}}} = \left(\frac{R_{15} R_8}{R_7 R_{12}} + \frac{R_{10} j\omega C_1 R_{15}}{(R_9 j\omega C_1 + 1) R_{13}} + \frac{R_{19} R_{15}}{R_{11} R_{14} (1 + j\omega C_2 R_{19})} \right) \tag{1.27}$$

With $s = j\omega$, $R_1 9 = \infty$ and

$$C(s) = K_p + \frac{K_d s}{1 + K_a s} + \frac{K_i}{s} \tag{1.28}$$

we can relate the resistors and capacitors to the controller gains by comparing to (1.27).

$$K_p = \frac{R_{15} R_8}{R_7 R_{12}} \tag{1.29}$$

$$K_d = \frac{R_{10} R_{15} C_1}{R_{13}} \tag{1.30}$$

$$K_a = C_1 R_9 \tag{1.31}$$

$$K_i = \frac{R_{15}}{R_{14} R_{11} C_2}, \tag{1.32}$$

e)

Tracking Control

With the currents

$$I_3 = \frac{U_{\text{ref1}}}{R_3}, \tag{1.33}$$
$$I_2 = \frac{U_{\text{err}}}{R_2} \tag{1.34}$$
$$\tag{1.35}$$

and Kirchhoff's Current Law,

$$I_2 + I_3 + I_4 = 0, \tag{1.36}$$

we derive

$$I_4 = -I_3 - I_2 = -\frac{U_{\text{ref1}}}{R_3} - \frac{U_{\text{err}}}{R_2}. \tag{1.37}$$

Using Ohm's law and rearranging yields:

$$U_{\text{out}}^{\text{tr}} = R_4 \cdot I_4 = -R_4 \left(\frac{U_{\text{ref1}}}{R_3} + \frac{U_{\text{err}}}{R_2} \right) \tag{1.38}$$

Under the assumption $U_{err} = 0$ the transfer function is derived:

$$\frac{U_{\text{out}}^{\text{tr}}}{U_{\text{ref1}}} = -\frac{R_4}{R_3} \tag{1.39}$$

Focus Control

With the transfer function of the tracking control already known, the transfer function for the focus control is quickly derived:

$$U_{\text{out}}^{\text{fo}} = U_{\text{out}}^{\text{tr}} \cdot \frac{R_6}{R_5} \tag{1.40}$$
$$U_{\text{out}}^{\text{fo}} = \frac{R_4 \cdot R_6}{R_5} \left(\frac{U_{\text{ref1}}}{R_3} + \frac{U_{\text{err}}}{R_2} \right) \tag{1.41}$$
$$\frac{U_{\text{out}}^{\text{fo}}}{U_{\text{ref1}}} = \frac{R_4 \cdot R_6}{R_5 \cdot R_3} \tag{1.42}$$

1.2 Part Two

To find the turnover frequencies, we rewrite

$$C(s) = K_p + \frac{K_d s}{1 + K_a s} + \frac{K_i}{s} \tag{1.43}$$

in a form, where we can directly read out the turnover frequencies:

$$C(s) = K_p + \frac{K_d s}{1+K_a s} + \frac{K_i}{s} \tag{1.44}$$

$$= \frac{K_p s(1+K_a s)}{s(1+K_a s)} + \frac{K_d s^2}{s(1+K_a s)} + \frac{K_i(1+K_a s)}{s(1+K_a s)} \tag{1.45}$$

$$= \frac{K_p s(1+K_a s) + K_d s^2 + K_i(1+K_a s)}{s(1+K_a s)} \tag{1.46}$$

$$= \frac{K_d s^2 + K_p K_a s^2 + K_p s + K_i + K_i K_a s}{s(1+K_a s)} \tag{1.47}$$

$$= \frac{K_i + (K_p + K_i K_a)\, s + (K_p K_a + K_d)\, s^2}{s(1+K_a s)} \tag{1.48}$$

$$= K_i \cdot \frac{1 + \left(\frac{K_p}{K_i} + K_a\right) s + \left(\frac{K_p K_a + K_d}{K_i}\right) s^2}{s(1+K_a s)} \tag{1.49}$$

$$= K_i \cdot \frac{1 + \left(\frac{K_p + K_i K_a}{K_i}\right) s + \left(\frac{K_p K_a + K_d}{K_i}\right) s^2}{s(1+K_a s)} \tag{1.50}$$

Rearranging from the generic factorized form with unknown turnover frequencies:

$$C(s) = \kappa \cdot \frac{\left(1 + \frac{s}{\omega_{e1}}\right)\left(1 + \frac{s}{\omega_{e2}}\right)}{s\left(1 + \frac{s}{\omega_{e3}}\right)} \tag{1.51}$$

$$= \kappa \cdot \frac{1 + \frac{s}{\omega_{e1}} + \frac{s}{\omega_{e2}} + \frac{s^2}{\omega_{e1}\omega_{e2}}}{s\left(1 + \frac{s}{\omega_{e3}}\right)} \tag{1.52}$$

$$= \kappa \cdot \frac{1 + \left(\frac{1}{\omega_{e1}} + \frac{1}{\omega_{e2}}\right) s + \frac{s^2}{\omega_{e1}\omega_{e2}}}{s\left(1 + \frac{s}{\omega_{e3}}\right)} \tag{1.53}$$

$$= \kappa \cdot \frac{\left(\frac{\omega_{e1}+\omega_{e2}}{\omega_{e1}\omega_{e2}}\right) s + \left(\frac{1}{\omega_{e1}\omega_{e2}}\right) s^2 + 1}{s\left(1 + \frac{s}{\omega_{e3}}\right)} \tag{1.54}$$

Now comparing (1.50) with (1.54), yields the following set of equations:

$$\omega_{e1}\omega_{e2} = \frac{K_i}{K_p K_a + K_d} \tag{1.55}$$

$$\omega_{e1} + \omega_{e2} = \frac{K_p + K_i K_a}{K_p K_a + K_d} \tag{1.56}$$

$$\omega_{e3} = \frac{1}{K_a} \tag{1.57}$$

Plugging in the definitions from (1.29) to (1.32) yields:

$$\omega_{e1}\omega_{e2} = \frac{\dfrac{R_{15}}{R_{14}R_{11}C_2}}{\dfrac{R_{15}R_8C_1R_9}{R_7R_{12}} + \dfrac{R_{10}R_{15}C_1}{R_{13}}} \tag{1.58}$$

$$\omega_{e1} + \omega_{e2} = \frac{\dfrac{R_{15}R_8}{R_7R_{12}} + \dfrac{R_{15}C_1R_9}{R_{14}R_{11}C_2}}{\dfrac{R_{15}R_8C_1R_9}{R_7R_{12}} + \dfrac{R_{10}R_{15}C_1}{R_{13}}} \tag{1.59}$$

$$\omega_{e3} = \frac{1}{C_1R_9} \tag{1.60}$$

With the rule of thumb

$$\omega_{e1} = 0.1\omega_c, \tag{1.61}$$
$$\omega_{e2} = 0.33\omega_c, \tag{1.62}$$
$$\omega_{e3} = 3.3\omega_c, \tag{1.63}$$
$$|C(j\omega_c)P(j\omega_c)| = 1 \tag{1.64}$$
$$= \left| \left(\frac{(K_pK_a + K_d)s^2 + (K_p + K_iK_a)s + K_i}{K_as^2 + s} \right) \cdot \frac{a}{s^2} \right|_{s=j\omega_c} \tag{1.65}$$

from the script, we can plug in values for the three turnover frequencies and add another equation. As the resistance values

$$R_{12} = 10\ \text{k}\Omega \tag{1.66}$$
$$R_{13} = 10\ \text{k}\Omega \tag{1.67}$$
$$R_{14} = 1\ \text{M}\Omega \tag{1.68}$$
$$R_{15} = 51\ \text{k}\Omega \tag{1.69}$$

are already given, we arrive at a nonlinear system of 4 equations with 4 unknown variables. This can now be solved numerically.

Experiments

1.3 System Identification

In this task, the system identification for the CD-Player is to be performed. Measuring the two frequency responses

$$F_1(s) = \frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)}, \tag{1.70}$$

$$F_2(s) = \frac{U(s)}{R(s)} = \frac{C(s)}{1 + C(s)P(s)}, \tag{1.71}$$

where $F_1(s)$ is the complementary sensitivity function $T(s)$ of the plant and

$$F_2(s) = C(s)S(s), \tag{1.72}$$

with the sensitivity function $S(s)$ of the plant. As

$$P(s) = \frac{T(s)}{S(s)C(s)} = \frac{F_1(s)}{F_2(s)}, \tag{1.73}$$

we can compute the plant transfer function $P(s)$ from our measured frequency responses. We perform this twice, once for the focus control loop and once for the tracking control loop.

1.3.1 Focus Control Plant Estimation

We measure both

$$F_1(s) = \text{FOREFtoFOER}, \tag{1.74}$$

$$F_2(s) = \text{FOREFtoFODR} \tag{1.75}$$

and then calculate

$$P(s) = \frac{F_1(s)}{F_s(s)}. \tag{1.76}$$

We can now fit a transfer function model to this data. As we calculated in (1.13) the transfer function of $P(s)$ has the simple form

$$P(s) = \frac{\alpha}{s^2}. \tag{1.77}$$

Fitting α to the data, yields $\alpha = 10364860.3963$. The data and fit is shown in figure 1.1

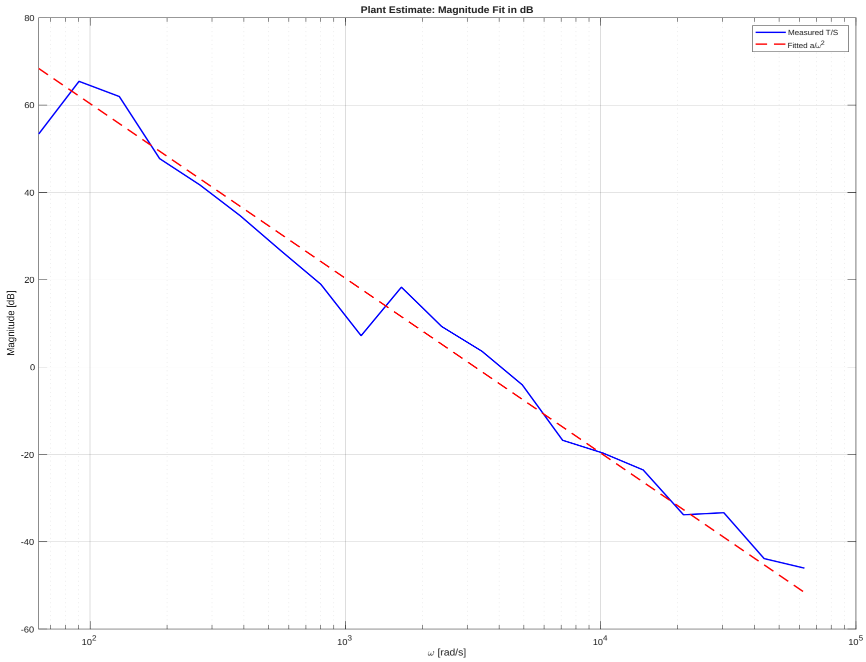

Figure 1.1: Magnitude of $P(s)$ for focus control and fitted alpha.

1.3.2 Tracking Control Plant Estimation

Similarly, we measure

$$F_1(s) = \text{TRREFtoTRER}, \tag{1.78}$$

$$F_2(s) = \text{TRREFtoTRDR} \tag{1.79}$$

and then calculate

$$P(s) = \frac{F_1(s)}{F_s(s)}, \tag{1.80}$$

for the tracking control. This yields the magnitude response given in 1.2, with a calculated value of $\alpha = 36335302.2753$.

1.4 Control Design

Using the nonlinear system of equations we derived in 1.2, together with the estimated α from 1.3 we can now numerically solve the nonlinear system of equations twice, once for the focus control and once for the tracking control. The calculated values are given in table 1.4. With these values for the resistors, we can now fully operate the CD player. After following steps $1-6$ in Section 5.3 of the script, we were able to fully operate the CD player. For validation, two Bode plots were measured.

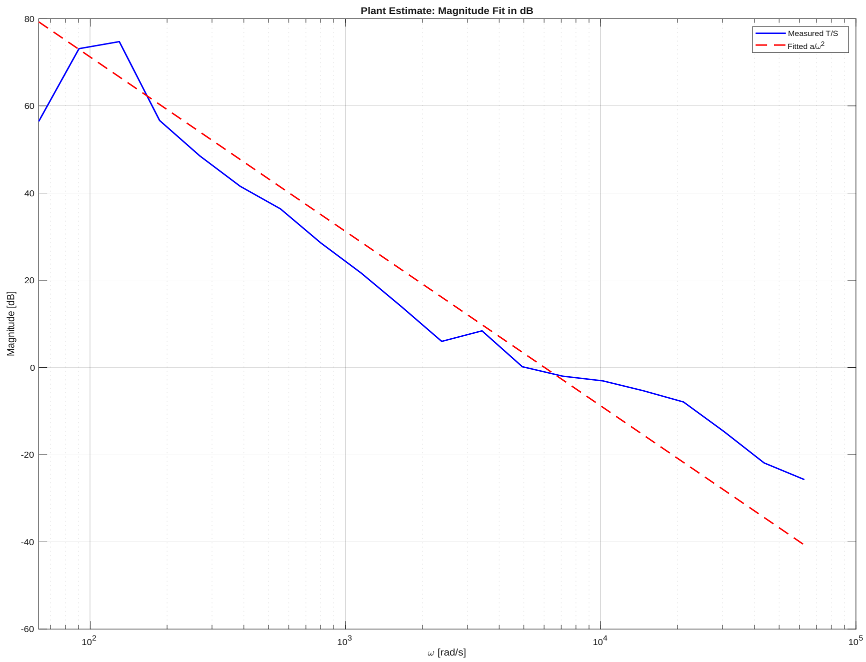

Figure 1.2: Magnitude of $P(s)$ for tracking control and fitted alpha.

Resistor	**Tracking Control**	**Focus Control**
R_8	0.43 kΩ	1.52 kΩ
R_9	2.09 kΩ	2.09 kΩ
R_{10}	0.62 kΩ	2.17 kΩ
R_{11}	1.42 kΩ	0.41 kΩ

Table 1.1: Comparison of resistor values for tracking and focus control

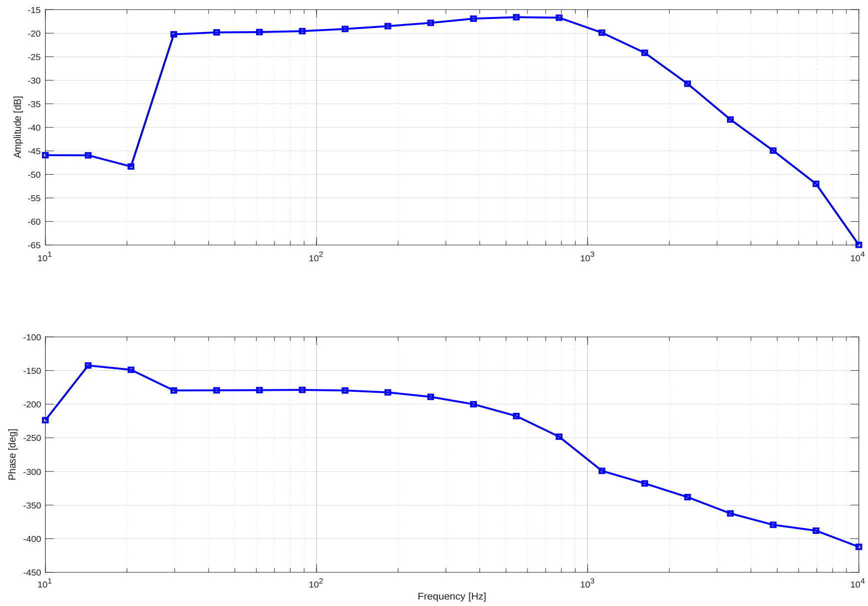

Figure 1.3: Bode plot from FOREF to FOER.

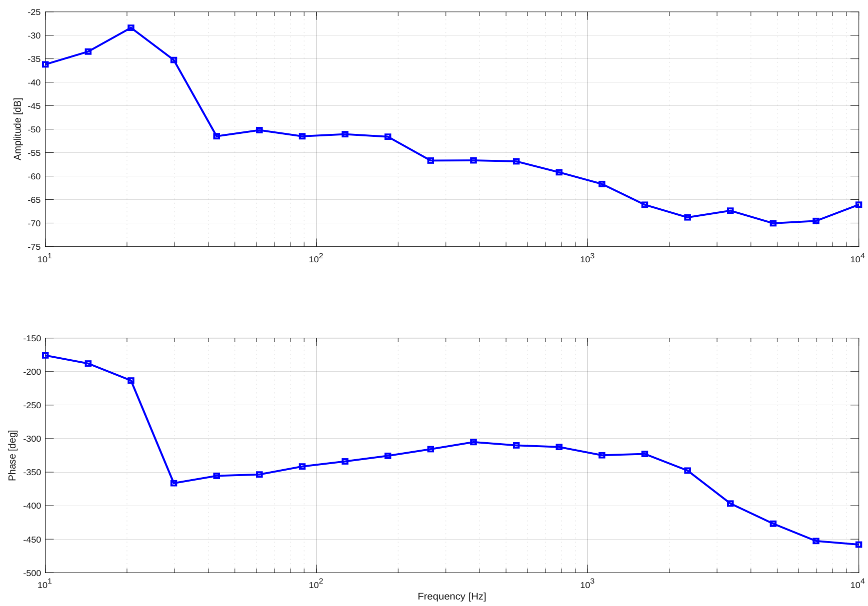

Figure 1.4: Bode plot from TRREF to TRER.

1.4.1 Focus Control Closed Loop Frequency Response

In figure 1.3, we can see the frequency response of the closed focus control loop. Looking at the magnitude response, we see that our system barely reacts to very low frequencies with an attenuation of −45 dB at 10 Hz. At about 25 Hz the magnitude jumps to about −20 dB and stays relatively flat until about 100 Hz. This is the active region of our controller. t is noted, however, that the magnitude response is still relatively low. Meaning our controller, while stable, may be unnecessarily conservative. The magnitude rolls off with about 40 dB/ Dekade starting around 100 Hz which is expected for a second-order-system. This roll-off also confirms that the controller effectively rejects high-frequency disturbances, which expected, as a low pass filter is used for the derivative action.

1.4.2 Tracking Control Closed Loop Frequency Response

In figure 1.4, we can see the frequency response of the closed tracking control loop. Unfortunately the magnitude response the system shows, is not really what we would expect for a well tuned PID-controlled second-order-system. The system shows significant attenuation of about −35 dB at 10 Hz. The magnitude of the response rises to about −28 dB at 20 Hz but then falls sharply to about −55 dB at 40 Hz The magnitude then continues to roll-off slowly with about 10 dB/ Dekade until 10 kHz. This suggests an overly conservative controller that, while stable, only reacts very slowly if at all to any disturbances.

1.4.3 CD Readout

With both controllers working, we can finally read out the data on the CD. For this, the CD is rotated while both controllers are turned on and stable. The output of the RF signal can be seen in 1.5. From the output, we can see a sinusoidal fluctuation of the RF signal. This can be explained by the pits and lands of the CD. Hence we can conclude that the CD is successfully read. The jitter and however, the unstable amplitude hints at not perfectly working controllers, which confirms the assumptions made from 1.3 and 1.4.

1.5 Report Questions

1.5.1 RF signal H and L level

Looking at figure 1.5, the "High Level" H of the signal, meaning the measured voltage, when the laser hits land, is about $V_{\mathrm{H}} = 560$ mV
The "Low Level" L of the signal, meaning the measured voltage, when the

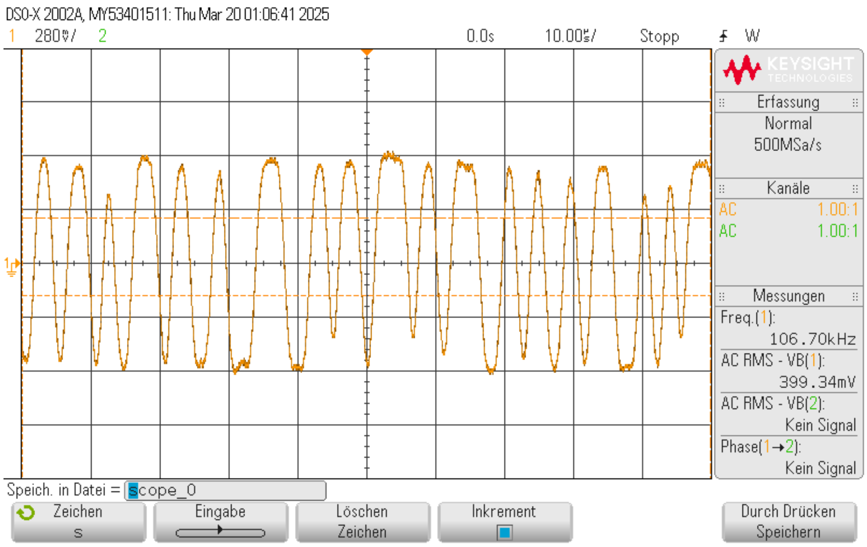

Figure 1.5: RF signal from reading out the CD

laser hits a pit, is about $V_{\mathrm{L}} = -560$ mV

1.5.2 Benefits of Integral Control

The main benefit of integrating integral action into any control loop is the elimination of the steady state error. A pure PI-controller cannot fully eliminate any constant offset or disturbance the system may exhibit.

1.5.3 Steady State Error under PID-Control

In theory the integral action of the PID-controller should eliminate the steady-state error. As alrady discussed, our PID controll loop did not work perfectly so there still is some offset. The offset could be reduced by increasing the integral gain of our controller.